How to Prove It

2.3 More operations on sets

Example 2.3.3 asks us to analyse the logical forms of several statements involving powersets.

§ **Example 2.3.3**

Parameters

+ A $\in$ Set
+ B $\in$ Set

Observations

1. $x \in \mathcal{P}\, A$

   $\equiv (\forall z \mid z \in x : z \in A)$

2. $\mathcal{P}\, A \subseteq \mathcal{P} B$

   $\equiv (\forall X \mid (\forall z \mid z \in X : z \in A) : (\forall z \mid z \in X : z \in B))$

3 $B \in \{\mathcal{P}\, A \mid A \in \mathcal{F}\}$

   $\equiv (\exists A \mid (\forall z \mid z \in A : z \in \mathcal{F}) : (\forall z \mid : z \in B \equiv (\forall y \mid y \in z : y \in A)))$

4. $x \in \mathcal{P}(A \cap B)$

   $\equiv x \in \mathcal{P}\, A \cap x \in \mathcal{P}\, B$

5. $\mathcal{P}(A \cap B)$

   $= \mathcal{P}\, A \cap \mathcal{P}\, B$

■